

Machine Learning-Driven Prediction of Emergency Department Length of Stay, Admission, and Hospitalization Using Data Available at Triage

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2 ABSTRACT

Background: Emergency Department (ED) crowding is a critical public health challenge. Prolonged ED length of stay (≥ 8 hours) affects over 20% of visits. Hospital admission and prolonged hospitalization (> 7 days) consume disproportionate resources. Clinicians currently rely on triage scales like the Emergency Severity Index (ESI), which have limited predictive accuracy. Machine learning (ML) using data available at triage could support these decisions, but evidence on model accuracy remains limited.

Methods: Using the MC-MED dataset (118,385 adult ED visits), we evaluated four ML models (XGBoost, LightGBM, Random Forest, TabNet) on three binary classification tasks using 29 variables available at or before the time of triage to predict: (1) prolonged ED length of stay (ED-LOS ≥ 8 hours vs. < 8 hours), (2) hospital admission (admit vs. discharge), and (3) prolonged hospital length of stay (hospital LOS > 7 days vs. ≤ 7 days) among admitted patients. Area under the receiver operating characteristic curve (AUC-ROC) was the primary performance metric. SHAP (SHapley Additive exPlanations) analysis identified the most important features for each prediction task.

Results: For ED-LOS ≥ 8 hours, XGBoost achieved an AUC-ROC of 0.659 (95% CI: 0.648–0.670). For hospital admission, XGBoost achieved an AUC-ROC of 0.816 (95% CI: 0.809–0.824), substantially outperforming the ESI baseline (AUC 0.505, 95% CI: 0.492–0.518). For hospital LOS > 7 days among admitted patients, Random Forest achieved an AUC-ROC of 0.670 (95% CI: 0.662–0.680). SHAP analysis identified consistent top predictors across tasks: triage acuity level, emergency medical services (EMS) arrival, age, disease condition count, and the Charlson Comorbidity Index.

Conclusion: ML using data at or before the time of triage shows strong discrimination for hospital admission and modest discrimination for predicting ED-LOS and prolonged hospital LOS. SHAP analysis provides interpretable feature importance to support clinical decision-making. External validation is required before clinical deployment.

Keywords: machine learning, emergency department, hospital admission, length of stay, Shapley Additive Explanations

1 INTRODUCTION

Emergency departments (EDs) offer a fundamental service, providing emergency treatments 24 hours a day, 365 days a year, therefore representing a critical public health service role [1, 2, 10]. Crowding in EDs has been recognised as a critical challenge in hospital administration [3, 4]. ED crowding has been linked to increased costs, longer waiting times, higher mortality rates, and adverse outcomes for vulnerable populations [6, 7, 8].

Emergency department length of stay (ED-LOS) is a significant indicator of ED bottlenecks. ED-LOS is defined as the interval from ED arrival to ED departure [9, 10]. A prolonged ED-LOS is a consequence of crowding and has been associated with a 37% elevated mortality risk among admitted patients [6]. Several studies have defined an extended ED-LOS as more than 3 hours [10], while others use thresholds of 4 to 12 hours [11]. Prolonged ED length of stay not only affects patient outcomes but also contributes to ambulance diversion, treatment delays for time-sensitive conditions, and inefficient resource utilization across the hospital system.

Clinicians currently rely on triage scales such as the Emergency Severity Index (ESI) to prioritize patients and predict clinical trajectories. However, the ESI demonstrates limited predictive accuracy for key outcomes such as hospital admission and prolonged length of stay when used in isolation, with AUC-ROC values typically below 0.70 [12]. This limitation creates an opportunity for data-driven approaches that integrate multiple patient characteristics available at triage.

Machine learning (ML) techniques are frequently used in healthcare for diagnosis, forecasting patient outcomes, and resource allocation [13, 14]. Following this approach, various studies have examined ways to predict ED-LOS, hospital admission, and hospital LOS using ML strategies [15, 16, 17]. Ricciardi et al. [10] achieved 74.8% accuracy for predicting ED-LOS greater than 3 hours using Random Forest. Wu et al. [15] found that LightGBM outperformed other models for continuous ED-LOS prediction in a Chinese hospital setting.

Shbool et al. [18] applied machine learning algorithms including Random Forest and Decision Trees to predict ED length of stay, achieving an accuracy of 72.8% using 15 triage variables on a dataset of 10,241 ED visits. They identified age, triage category, and mode of arrival as the most important predictors. Hong et al. [19] demonstrated that machine learning models using triage data, demographics, and patient history can predict hospital admission with AUC up to 0.88 using gradient boosting. Their study of 329,000 ED visits showed that the ESI alone had limited predictive power (AUC 0.68), but machine learning models incorporating additional triage variables significantly improved performance.

However, existing models have three critical limitations. First, most studies predict ED-LOS, hospital admission, or hospital LOS in isolation, despite their clinical interdependence in real-world emergency care pathways [9]. Second, many models incorporate laboratory results or specialist consultations that are unavailable at triage, limiting real-time clinical applicability [12]. Third, while SHAP has been applied to individual ED prediction tasks, its systematic application across multiple cascading outcomes (ED-LOS, admission, and hospital LOS) using a unified triage-time feature set has not been explored.

Given these considerations, this work aims to build a predictive model to forecast three binary treatment outcomes using only variables available at or before triage: (1) prolonged ED-LOS (≥ 8 hours), (2) hospital admission, and (3) prolonged hospital LOS (> 7 days) among admitted patients.

Using the MC-MED dataset (118,385 adult ED visits from Stanford Health Care, 2020 to 2022), we evaluated four ML algorithms (XGBoost, LightGBM, Random Forest, and TabNet) on three binary classification tasks using 29 variables available at triage. Predicting these outcomes in advance would be beneficial for planning staff allocation and bed management with the goal of reducing ED crowding.

2 METHODS

2.1 Prediction Tasks

Three binary classification tasks were formulated using only variables available at triage. Task 1 predicted prolonged emergency department length of stay defined as eight hours or more versus less than eight hours [6, 11]. The ED-LOS threshold of 8 hours was selected based on evidence linking prolonged ED boarding beyond this duration to elevated mortality (Sun et al., 2013) and its use as a standard crowding benchmark.

Task 2 predicted hospital admission, where admission was defined as any non-discharge disposition, including inpatient, observation, and intensive care unit admissions. Discharge to home, transfer to another facility, death in the ED, hospice discharge, and leaving against medical advice (LAMA) were all classified as non-admission (discharge) outcomes [57, 16].

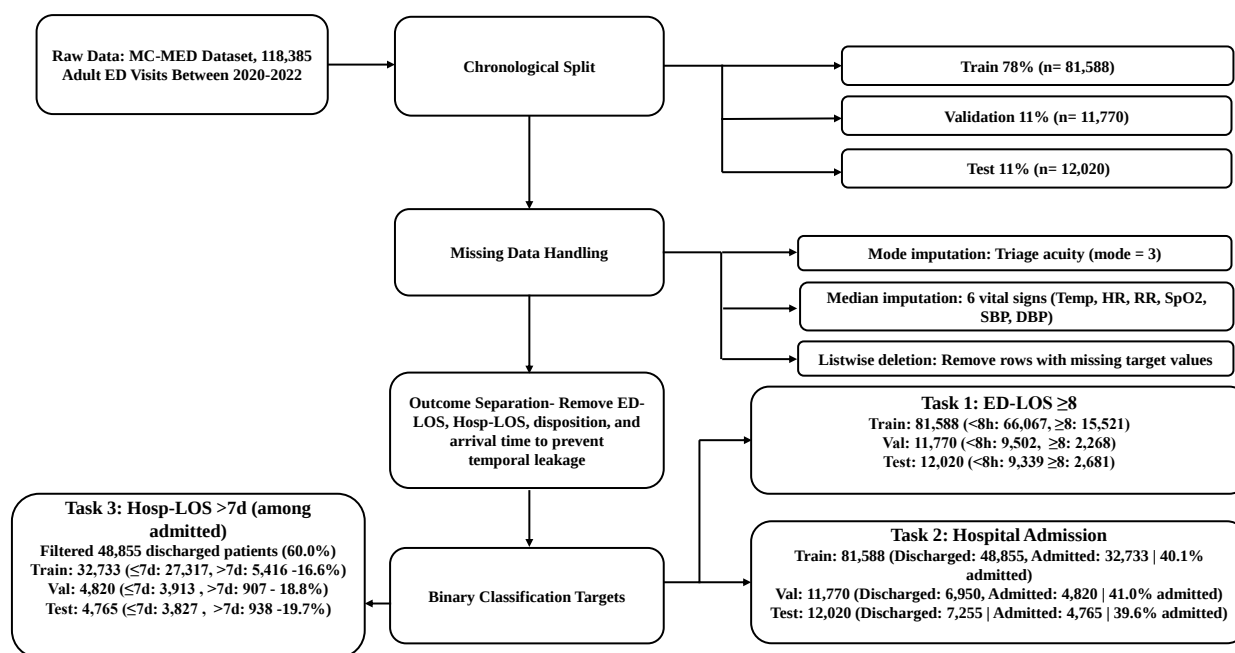


Figure 1. Chronological preprocessing pipeline for the MC-MED dataset showing split proportions, missing data handling methods (median, mode), outcome separation, and three binary classification targets.

Task 3 predicted prolonged hospital length of stay defined as more than seven days versus seven days or less, applied exclusively to admitted patients [9, 63]. The hospital LOS threshold of 7 days was chosen to identify patients with complex inpatient courses requiring extended hospitalization, consistent with prior systematic reviews [63, 9].

ED-LOS was calculated as the interval (in hours) from ED Arrival.time to Departure.time from the emergency department. Hospital LOS was calculated as the interval (in days) from hospital admission time (Admit.time) to hospital discharge time, among admitted patients only. Patients with missing Admit.time or discharge timestamps were excluded from Task 3 via listwise deletion.

2.2 Data Resource and Preprocessing

This study analysed the MC-MED dataset [57, 58], which contains 118,385 adult emergency department visits from Stanford Health Care between January 2020 and December 2022. The dataset includes demographics, triage vital signs, chief complaints, comorbidity measures, home medications, and visit outcomes [59].

Figure 1 shows the data preprocessing pipeline. The pipeline consisted of five sequential stages. Stage one involved raw data loading from the three source tables provided with the MC-MED dataset: visits.csv, meds.csv, and pmh.csv. These tables were merged using the Contact Serial Number to link visit specific records and the Medical Record Number (MRN) to aggregate longitudinal medication and past medical history data. Stage two applied the chronological split into training, validation, and test sets using the pre-defined split files provided with MC-MED, which ensured temporal separation between periods with no patient appearing in more than one split. Stage three handled missing data using the procedures described in detail below. Stage four performed outcome separation, where all time dependent columns including emergency department length of stay, hospital length of stay, disposition status, arrival time, admit time, discharge time, and departure time were removed from the feature set before model training to prevent temporal and data leakage. Stage five defined the three binary classification targets as described in the preceding subsection.

2.2.1 Missing Data Handling

Missing data were handled using three distinct strategies applied sequentially, with all imputation parameters derived exclusively from the training set to prevent information leakage from the validation or test sets.

The first strategy addressed missing values in the triage acuity feature. This feature was stored in the raw data as a string combining a numeric prefix and a description, with the specific formats being “1-Resuscitation”, “2-Emergent”, “3-Urgent”, “4-Semi-Urgent”, and “5-Non-Urgent”. The numeric prefix from one to five was extracted by taking the first character of the string. Missing values in this feature, which affected 8.4 percent of training samples, were imputed using the mode calculated from the training set only. The mode value was three, corresponding to the “3-Urgent” triage level. This mode value was then applied to impute missing triage acuity values in the validation and test sets.

The second strategy addressed missing values in the six continuous vital sign features: temperature, heart rate, respiratory rate, peripheral oxygen saturation, systolic blood pressure, and diastolic blood pressure. Missingness rates for these features ranged from 0.1 percent to 8.4 percent of training samples. These missing values were imputed using feature-wise median values. The SimpleImputer class from the scikit-learn library [60] was employed with the strategy parameter set to ‘median’. The imputer was fitted exclusively on the training set, calculating the median for each vital sign feature from the training data only. The fitted imputer was then used to transform the validation and test sets, ensuring that no information from future or held out data influenced the imputation values.

The third strategy handled missing target values. For each prediction task, any sample with a missing target value was removed from the analysis using listwise deletion. For Task 3, which predicted prolonged hospital length of stay among admitted patients only, this resulted in the exclusion of 142 training samples where hospital length of stay was not recorded.

The absence of medication or diagnosis records in `pmh.csv` or `meds.csv` for a given MRN was treated as true zero (i.e., no documented medications or conditions), not as missing data. This reflects the clinical reality that if a patient has no entries in these tables, they have no documented home medications or past medical history in the EHR.

After applying these three strategies, no missing values remained in the feature set. All 29 features were complete for all samples across the training, validation, and test splits. This complete data matrix enabled direct compatibility with all four machine learning algorithms, including TabNet which cannot process missing values natively.

The dataset was chronologically split using the pre-defined split files provided with MC-MED, which ensure temporal separation between training, validation, and test periods [57]. A total of 13,007 patient encounters were excluded from the pure chronological ordering because their Contact Serial Number was not present in the chronological split files. The final analytical cohort comprised 105,378 visits.

Table 1 presents the demographic and clinical characteristics of the study cohort across the three chronological splits. The mean age was 52.3 years with a standard deviation of 19.8 years, with female patients accounting for 52.1 percent of visits. The baseline admission rate was 40.1 percent in the analytical cohort. The prevalence of prolonged emergency department length of stay, defined as eight hours or more, was 22.3 percent. The prevalence of prolonged hospital length of stay, defined as more than seven days among admitted patients, was 19.7 percent.

2.3 Feature Set and Temporal Integrity

A total of 29 variables available at or before triage were extracted for each emergency department visit by integrating the three source tables. The resulting predictors were organized into six clinical domains: demographics, arrival context, triage vitals and acuity, chief complaint, comorbidity burden, and medication burden. All variables represent the patient’s clinical state at or before the moment of ED arrival. Demographics, triage vital signs, triage acuity, and chief complaint are captured during the triage process (typically within 30–90 minutes of arrival). Longitudinal features (past medical history, home medications, Charlson Index) are derived from historical records documented prior to the index visit.

Table 1. Descriptive characteristics of the MC-MED analytical cohort by chronological split

Characteristic	Full Analytical Cohort	Train (77.4%)	Val (11.2%)	Test (11.4%)
N	105,378	81,588	11,770	12,020
Age, mean (SD)	52.3 (19.8)	52.1 (19.7)	52.5 (19.9)	52.4 (19.8)
Female (%)	52.1	52.0	52.2	52.1
Race (% White)	62.3	62.2	62.4	62.3
EMS arrival (%)	31.7	31.6	31.9	31.7
Admission rate (%)	40.1	40.1	41.0	39.6
ED-LOS 8h or more (%)	22.3	22.1	22.5	22.3
Hosp-LOS more than 7d (%)*	19.7	19.6	19.9	19.7
Charlson Index, mean (SD)	1.8 (1.9)	1.8 (1.9)	1.8 (1.9)	1.8 (1.9)

*Among admitted patients only: total n=42,318 (Train=32,733, Val=4,820, Test=4,765)

To strictly prevent temporal leakage, the visit-level records were linked to the patient-level medication and past medical history tables using the Medical Record Number (MRN). This analysis utilized the official MC-MED chronological split [57], which restricts each patient to only one of the training, validation, or test sets, preventing cross-split leakage. Furthermore, to prevent leakage from subsequent visits within the same split, the longitudinal aggregation was constrained by the index visit timestamp: only diagnoses (`Noted_date`) and medication orders (`Entry_date`) occurring on or before the index ED Arrival_time were included. This ensures that home medication counts and comorbidity burdens reflect only the patient's true pre-arrival clinical state.

Home medication features were derived from the `meds.csv` table after temporal filtering (`Entry_date` $\leq$ index Arrival_time). The 'Home Medication Count' represents the number of unique medication entries (by Med_ID) documented prior to the index visit. 'Unique Medication Classes Count' represents the number of distinct Med_class categories among these pre-arrival medications. The 'Polypharmacy Flag' is set to 1 if the Home Medication Count ≥ 5 . Duplicate medication entries (same Med_ID for the same patient) were retained as separate rows if they had distinct Entry_dates, reflecting repeated prescriptions over time; however, for the purposes of the polypharmacy flag, we counted unique Med_IDs to avoid inflating counts from duplicate entries.

The Charlson Comorbidity Index and Disease Condition Count were calculated exclusively from diagnoses in `pmh.csv` with `Noted_date` $\leq$ index Arrival_time. The Charlson Index uses the Quan et al. [62] ICD-10 mapping algorithm applied to these pre-arrival diagnoses. The Disease Condition Count represents the number of unique CCS (Clinical Classifications Software) categories among these pre-arrival diagnoses.

To prevent temporal and data leakage, no post-triage or outcome-dependent variables including emergency department length of stay, hospital length of stay, disposition status, or post-admission timestamps were included in the predictor set. Table 2 provides the complete feature list with data nature and derivation method.

2.4 Machine Learning Algorithms

Four machine learning algorithms were implemented: Random Forest, XGBoost, LightGBM, and TabNet. These algorithms follow different theoretical approaches [10, 15]. Random Forest [20] uses bootstrapping to train numerous decision trees concurrently and combines their outputs using bagging [10]. XGBoost [21] is a gradient boosting algorithm with regularization to prevent overfitting [56]. LightGBM [22] is a histogram based gradient boosting algorithm optimized for efficiency [15]. TabNet [23] is a deep learning architecture with attention mechanisms specifically designed for tabular data [17].

2.4.1 Hyperparameter Selection

Hyperparameters for each model were optimized separately for each of the three prediction tasks using Bayesian search with ten-fold cross-validation on the training set, targeting the area under the receiver operating characteristic curve metric. The search space for each model included key parameters such as

tree depth, learning rate, and the number of estimators, as detailed in Table 3. The best hyperparameter configuration was selected based on the mean validation AUC-ROC averaged across the ten folds.

To address class imbalance, `scale_pos_weight` (for XGBoost) was calculated separately for each task based on the training set class distribution:

- Task 1 (ED-LOS ≥ 8 h): `scale_pos_weight` = 4.26 (prevalence 19.0%)
- Task 2 (Hospital Admission): `scale_pos_weight` = 1.49 (prevalence 40.1%)
- Task 3 (Hosp-LOS > 7 d): `scale_pos_weight` = 5.04 (prevalence 16.5%)

Final model performance was then evaluated on the held-out test set.

2.4.2 Cross Validation Procedure

Ten-fold cross-validation was performed on the training set of 81,588 patients to ensure reliable performance estimates and prevent overfitting. The training data were partitioned into ten approximately equal-sized folds while preserving patient-level separation, meaning no patient (MRN) appeared in more than one fold. For each fold, the model was trained on nine folds and validated on the remaining fold. This process was repeated ten times, with each fold serving as the validation set once. The reported cross-validation metrics represent the mean values across the ten folds along with their standard deviations. After hyperparameter optimization, each model was retrained on the full training set using the optimal hyperparameters and evaluated on the independent test set.

2.4.3 Evaluation Metrics

The quality of the produced models was assessed using standard classification metrics. Let TP represent true positives, TN represent true negatives, FP represent false positives, and FN represent false negatives. We computed accuracy, precision, sensitivity (recall), specificity, F1-score, and AUC-ROC.

AUC-ROC evaluates the model's capacity to distinguish between positive and negative classes across all classification thresholds. Probability thresholds for classification metrics (sensitivity, specificity, accuracy, precision) were selected by maximizing Youden's J statistic ($\text{Sensitivity} + \text{Specificity} - 1$) on the independent validation set to prevent optimistic bias from threshold tuning on the test set. The optimal thresholds were:

- ED-LOS ≥ 8 h (XGBoost): 0.421
- Hospital Admission (XGBoost): 0.398
- Hosp-LOS > 7 d (Random Forest): 0.412

To assess model utility beyond discrimination, we report Brier scores, Precision-Recall AUC (AUC-PR), calibration intercept and slope, and threshold-based performance at actionable risk levels (Top 10% and Top 20% predicted risk).

Confidence intervals for all performance metrics were calculated using 1,000-iteration non-parametric bootstrapping on the independent test set, with percentile-based 95% CIs.

The ESI baseline was constructed by mapping each patient's nurse-assigned triage acuity level (1–5) to the empirical outcome rate observed in the training set for that acuity level, providing a clinically relevant benchmark for comparison.

All analyses were conducted using Python version 3.13 with the scikit-learn (v1.5.0), XGBoost (v2.1.0), LightGBM (v4.4.0), PyTorch (v2.3.0), and SHAP (v0.45.1) libraries. A fixed random seed (seed=42) was used for complete reproducibility.

2.5 Interpretability: SHAP Analysis

SHAP values (SHapley Additive exPlanations) [24] were computed to identify the most important features for each prediction task [61, 55]. SHAP analysis was conducted using the best performing model for each task. For tree-based models (XGBoost, LightGBM, Random Forest), we used TreeExplainer with the 'interventional' feature perturbation method, which computes exact SHAP values in polynomial time. For TabNet, we used KernelExplainer with 100 background samples (randomly selected from the training

set) to approximate SHAP values. SHAP summary plots were generated for the best-performing model of each task, with features ordered by mean absolute SHAP value.

SHAP values indicate the magnitude and direction of a feature's contribution to the model's output but do not imply causal relationships. Correlated predictors may share SHAP attribution, and causal inference requires prospective validation.

2.6 Sensitivity Analyses

To assess model robustness across patient subgroups, we conducted stratified sensitivity analyses. The test set was partitioned by three clinically relevant stratifiers: chief complaint (derived from the seven CC variables in Table 2), age group (<65 , $65-74$, $75-84$, ≥ 85 years), and triage acuity level (Acuity 1 through 5, with Acuity 4 and 5 combined due to small sample sizes). For each subgroup, we calculated AUC-ROC with 95% confidence intervals using 1,000 bootstrap resamples.

To assess fairness, we evaluated model performance across demographic subgroups including sex and race. Additionally, we performed a sensitivity analysis excluding triage acuity from the feature set to quantify its contribution to model performance relative to the underlying patient variables.

3 RESULTS

3.1 Task Performance

Table 4 presents the evaluation metrics for all four ML algorithms across the three prediction tasks. XGBoost achieved the best performance for ED-LOS and hospital admission prediction, while Random Forest performed best for hospital LOS prediction among admitted patients.

For ED-LOS ≥ 8 hours, XGBoost achieved an AUC-ROC of 0.659 (95% CI: 0.648–0.670). At the optimal threshold (0.421, selected on the validation set), the model achieved a sensitivity of 71.6% and specificity of 51.8%.

For hospital admission, XGBoost achieved an AUC-ROC of 0.816 (95% CI: 0.809–0.824), substantially outperforming the ESI baseline (AUC 0.505, 95% CI: 0.492–0.518). At the optimal threshold (0.398), the model achieved a sensitivity of 62.6% and specificity of 70.6%.

For hospital LOS >7 days among admitted patients, Random Forest achieved an AUC-ROC of 0.670 (95% CI: 0.662–0.680). At the optimal threshold (0.412), the model achieved a sensitivity of 62.3% and specificity of 72.0%.

XGBoost and LightGBM achieved statistically equivalent performance for hospital admission (DeLong test $p=0.42$).

3.2 Calibration and Clinical Utility

For ED-LOS prediction, XGBoost achieved a Brier score of 0.194 (AUC-PR: 0.312). Calibration analysis revealed an intercept of -1.136 and a slope of 0.693, indicating that while the model ranks patients well, the raw predicted probabilities are somewhat extreme. This highlights a critical operational requirement: the models require probability recalibration (e.g., via isotonic regression or Platt scaling) before raw outputs can be used as true absolute risk estimates.

To provide actionable insights for ED operations, we evaluated model performance at specific high-risk thresholds. For the XGBoost model at the Top 10% risk threshold (≥ 0.622): sensitivity = 19.6%, specificity = 92.3%, PPV = 38.6%. At the Top 20% risk threshold (≥ 0.543): sensitivity = 34.2%, specificity = 83.5%, PPV = 33.7%. This identifies a highly enriched subset where approximately one in three flagged patients will experience prolonged ED-LOS.

3.3 Subgroup Sensitivity Analyses and Fairness

Table 5 presents model performance stratified by chief complaint, age group, and triage acuity level. Hospital admission prediction remained stable across all subgroups, with AUC values ranging from 0.704 (patients aged ≥ 85 years) to 0.862 (fever complaints). In contrast, ED-LOS prediction showed declining discrimination with increasing age, from 0.688 in patients under 65 years to 0.582 in patients aged ≥ 85 .

years. Performance for prolonged hospital LOS among admitted patients was broadly consistent across subgroups, though small sample sizes in the highest acuity category (Acuity 1, $n=33$) warrant cautious interpretation.

Chief complaint subgroups are not mutually exclusive; patients with multiple complaints appear in each relevant subgroup. Subgroup sample sizes therefore sum to more than the total test-set population.

Fairness Analysis: ED-LOS prediction performance was relatively equitable across demographic subgroups. For sex, female patients ($n=6,411$) showed AUC 0.671 [0.656–0.686] and male patients ($n=5,602$) showed AUC 0.652 [0.636–0.668]. For race, White ($n=4,905$) AUC 0.645 [0.630–0.660], Black ($n=591$) AUC 0.615 [0.584–0.646], Asian ($n=2,234$) AUC 0.688 [0.670–0.706], and Other ($n=4,234$) AUC 0.676 [0.663–0.689].

Sensitivity Analysis (Triage Acuity Removal): Excluding triage acuity from the feature set reduced AUC by 0.023–0.038 across tasks. ED-LOS AUC fell from 0.659 to 0.621, Hospital Admission AUC fell from 0.816 to 0.789, and Hosp-LOS AUC fell from 0.670 to 0.648. This confirms that triage acuity contributes meaningful predictive information but that the models also learn from the underlying patient variables (vitals, demographics, comorbidities) rather than simply replicating the nurse's assessment.

3.4 SHAP Analysis

Figure 2 presents the SHAP summary plot for XGBoost for ED-LOS ≥ 8 h prediction. The top five features by mean absolute SHAP value were: EMS arrival, age, triage acuity level, disease condition count, and home medication count. Higher EMS arrival values (red) were associated with increased predicted risk of prolonged ED length of stay, consistent with EMS-transported patients having higher acuity and requiring more resources before disposition.

Figure 3 presents the SHAP summary plot for XGBoost for hospital admission prediction. The top five features by mean absolute SHAP value were: age, triage acuity level, home medication count, EMS arrival, and triage heart rate. Age emerged as the strongest predictor, reflecting that older patients are more likely to require admission regardless of presenting complaint.

Figure 4 presents the SHAP summary plot for Random Forest for hospital LOS > 7 d prediction among admitted patients. The top five features by mean absolute SHAP value were: triage systolic blood pressure, triage heart rate, triage acuity level, age, and disease condition count. Vital signs played a larger role here, indicating that physiological instability at triage is an important predictor of prolonged hospitalization.

Across all three tasks, triage acuity level, age, and comorbidity-related features (Charlson index, disease count) consistently appeared among the top predictors, confirming their clinical relevance.

Beyond these common predictors, each task had unique top features that reflect different stages of patient care. For ED-LOS prediction, EMS arrival emerged as a key predictor, suggesting that patients requiring ambulance transport face longer ED length of stay due to higher acuity and complex care coordination. For hospital admission prediction, triage heart rate was uniquely important, indicating that physiological stress at presentation is strongly associated with disposition decisions. For prolonged hospitalization prediction among admitted patients, triage systolic blood pressure and triage heart rate were important predictors, suggesting that vital sign abnormalities at triage are associated with complex inpatient courses requiring extended hospital stays. These differences suggest that while baseline demographic and comorbidity factors are universally important, physiological and logistical factors become more relevant at specific decision points.

4 DISCUSSION

ED-LOS represents a crucial key indicator of the efficiency of healthcare services. The ability to understand the reasons for prolonged ED-LOS could support the detection of bottlenecks in ED organisation. This study demonstrates that machine learning using data available at triage can predict three critical ED outcomes, with strong performance for hospital admission and modest performance for ED-LOS and prolonged hospitalization.

The evaluation metrics showed that XGBoost and Random Forest consistently outperformed the other algorithms. For ED-LOS prediction, XGBoost achieved an AUC-ROC of 0.659. For hospital admission

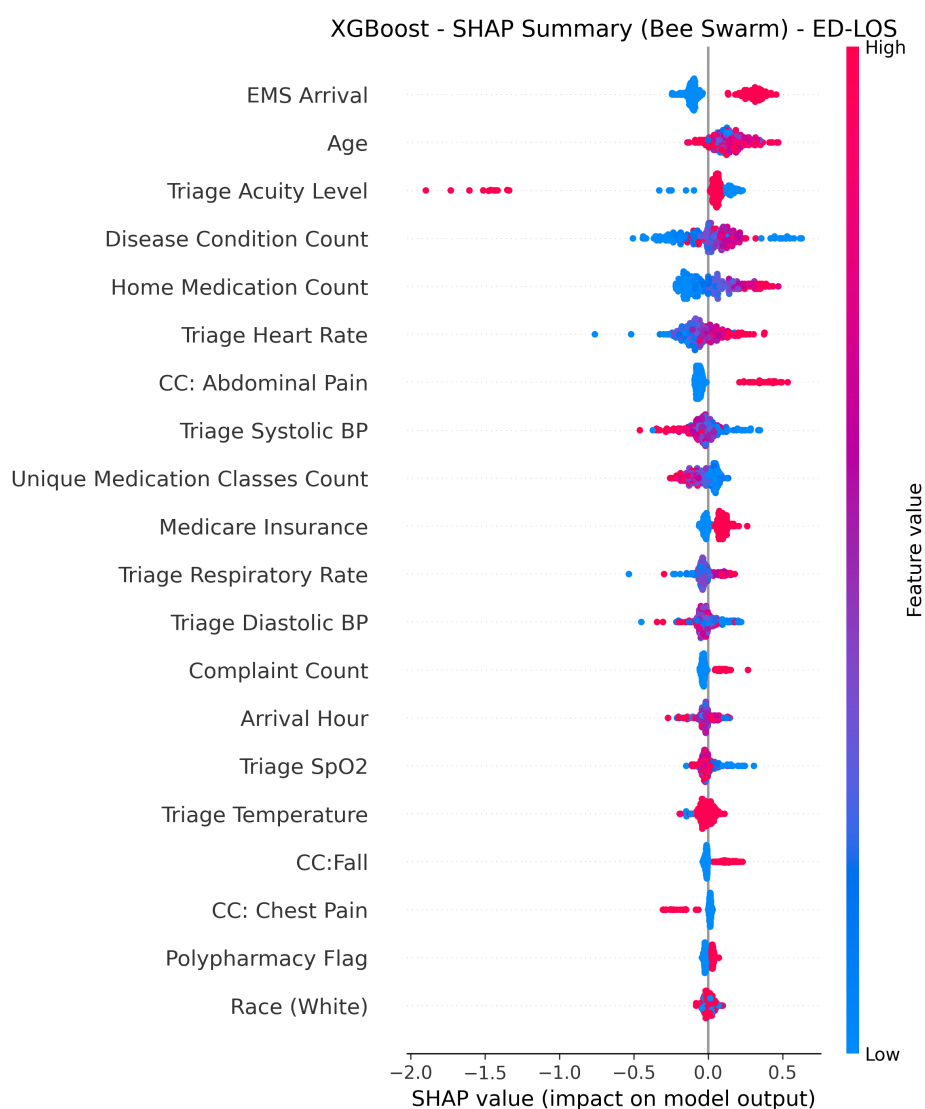


Figure 2. SHAP summary plot for XGBoost for ED-LOS ≥ 8 h prediction. Features are ordered by mean absolute SHAP value. Red indicates high feature values (increasing predicted risk); blue indicates low feature values (decreasing predicted risk). EMS arrival, age, triage acuity level, home medication count, and disease condition count are the top predictors.

prediction, XGBoost achieved an AUC-ROC of 0.816, substantially outperforming the ESI baseline (AUC 0.505). For hospital LOS prediction among admitted patients, Random Forest achieved an AUC-ROC of 0.670.

Comparing with other ML-based studies, Ricciardi et al. [10] achieved AUC approximately 0.68 for predicting ED-LOS greater than 3 hours using Random Forest. Our ED-LOS model (AUC 0.659) shows comparable discrimination for a more stringent threshold of 8 hours. Hong et al. [19] reported AUC up to 0.88 for admission prediction, and our admission model (AUC 0.816) is consistent with their findings for ML models using triage data. Shbool et al. [18] achieved accuracy of 72.8% for ED-LOS prediction; however, direct comparison is limited by differences in thresholds, populations, and evaluation procedures. Naemi et al. [27] achieved accuracy ranging from 75% to 82% for ED-LOS prediction. Our results are consistent with these findings.

A key contribution of this study is the systematic SHAP analysis across all three prediction tasks [55, 56]. Triage acuity level, age, and comorbidity burden were consistently important across all three tasks. Beyond these common predictors, task-specific features emerged. EMS arrival was distinctively

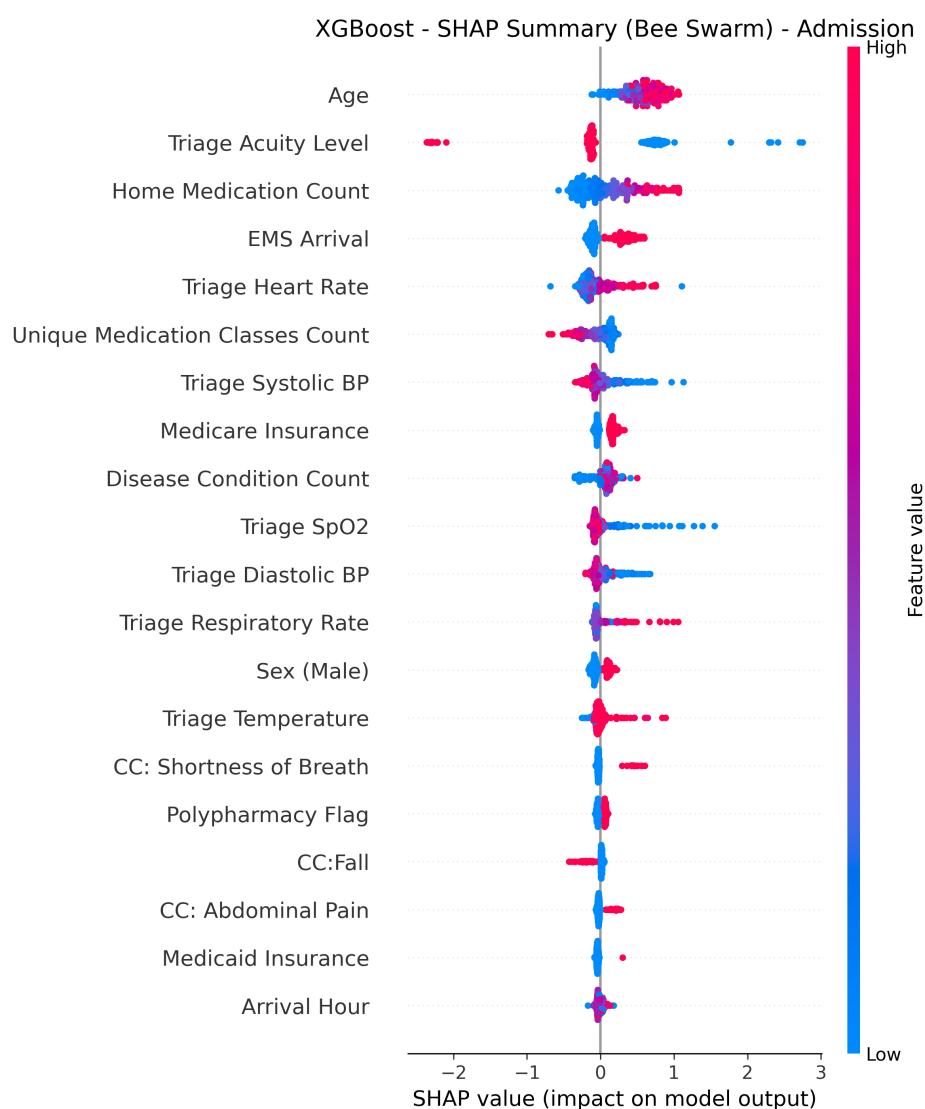


Figure 3. SHAP summary plot for XGBoost for hospital admission prediction. Features are ordered by mean absolute SHAP value. Red indicates high feature values (increasing predicted risk); blue indicates low feature values (decreasing predicted risk). Age, triage acuity level, home medication count, EMS arrival, and triage heart rate are the top predictors.

346 associated with prolonged ED length of stay, reflecting that ambulance-transported patients require
 347 more resources before disposition. Triage heart rate was strongly associated with admission predictions,
 348 indicating that physiological stress drives disposition choices. Vital signs (systolic blood pressure and
 349 heart rate) were important predictors of prolonged hospitalization, suggesting that physiological instability
 350 at triage is associated with complex inpatient courses. These findings demonstrate how different clinical
 351 factors become relevant at different stages of patient care.

352 Subgroup sensitivity analyses revealed stable hospital admission performance across chief complaints
 353 (AUC range: 0.704–0.862) and triage acuity levels. ED-LOS prediction showed declining discrimination
 354 with increasing age (AUC 0.688 in patients <65 years to 0.582 in patients ≥85 years), suggesting that the
 355 model performs less well in the oldest patients, who may have more complex clinical trajectories. Fairness
 356 analysis demonstrated relatively equitable performance across sex and race subgroups.

357 Missing data were handled using median imputation for continuous vital signs and mode imputation for
 358 categorical variables (e.g., triage acuity), based strictly on training set distributions to prevent data leakage.
 359 This ensures complete consistency with Section 2.2.1 (Methods) and our publicly available analysis code.

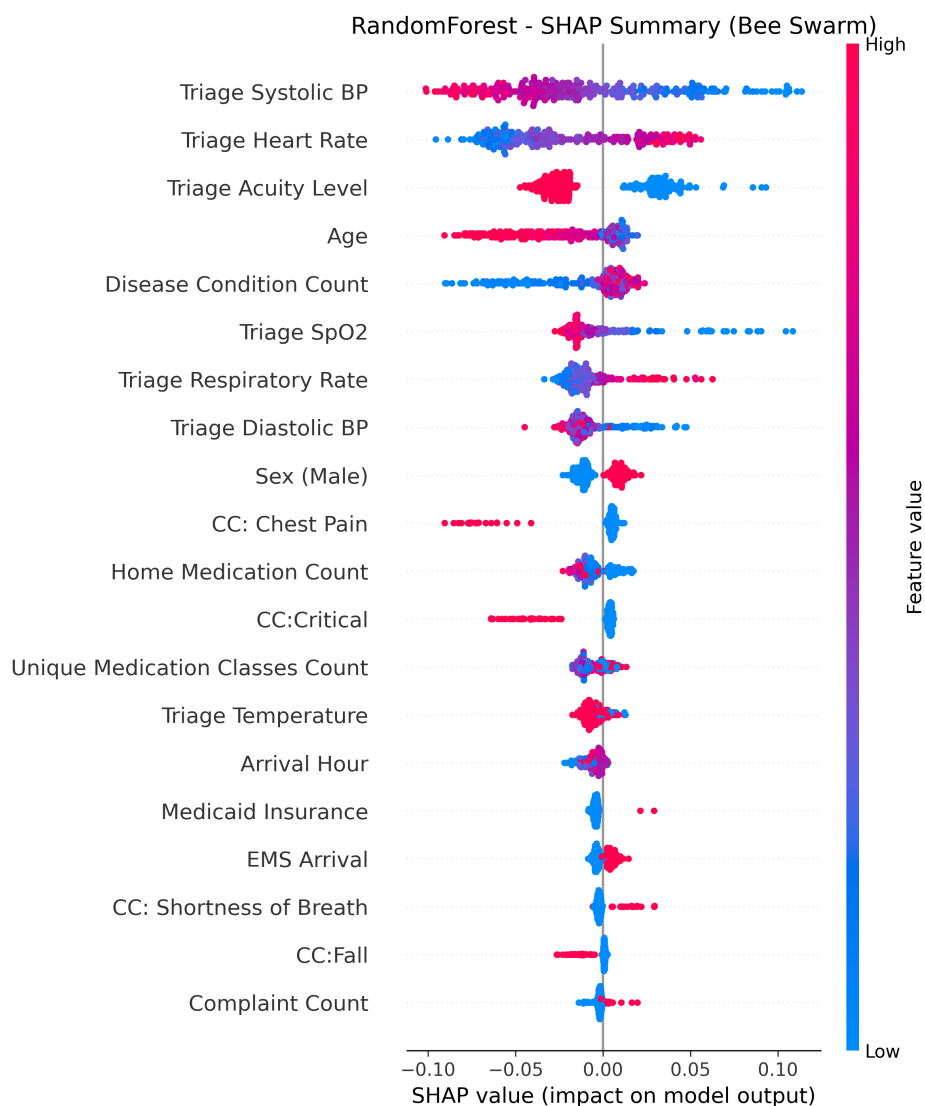


Figure 4. SHAP summary plot for Random Forest for hospital LOS >7 days prediction among admitted patients. Features are ordered by mean absolute SHAP value. Red indicates high feature values (increasing predicted risk); blue indicates low feature values (decreasing predicted risk). Triage systolic blood pressure, triage heart rate, triage acuity level, age, and disease condition count are the top predictors.

360 Calibration analysis revealed that while the models rank patients well, the raw predicted probabilities
 361 require recalibration before clinical deployment. The calibration slopes deviating from 1.0 (ED-LOS
 362 XGBoost: slope 0.693) indicate that probability calibration via isotonic regression or Platt scaling would
 363 be necessary for use as absolute risk estimates.

364 Several important determinants of ED-LOS are not captured in our dataset, including inpatient boarding
 365 time, bed availability, staffing levels, laboratory turnaround time, and imaging delays. These operational
 366 factors are well-established drivers of ED crowding and prolonged length of stay [25, 26]. Our triage-time
 367 model necessarily cannot incorporate these downstream operational variables, which represents both a
 368 limitation and an opportunity: our predictions reflect the patient's intrinsic clinical complexity at arrival,
 369 while operational factors represent modifiable system-level contributors. Future work integrating real-
 370 time operational data (e.g., current ED census, inpatient bed availability) with our clinical predictions
 371 could yield even more accurate ED-LOS forecasts.

372 The results of this study could inform future prospective studies evaluating ED management
 373 optimization. By predicting prolonged ED-LOS, decision makers could potentially allocate additional

374 medical and nursing staff during critical time slots. By predicting hospital admission, bed management
 375 could be improved. By predicting prolonged hospital LOS, discharge planning could be initiated earlier.
 376 However, these benefits require prospective validation.

377 4.1 Limitations

378 This study has several limitations that should be acknowledged. First, it is a single-centre retrospective
 379 study using data from one academic medical centre in California during 2020–2022, which includes the
 380 COVID-19 pandemic period when ED operations and patient acuity differed substantially from baseline.
 381 This limits generalizability to community hospitals, rural settings, and international healthcare systems
 382 [66, 67]. Variables such as Medicaid insurance are US-specific, and chief complaint categorisation may
 383 differ across settings. External validation across diverse healthcare systems is necessary before widespread
 384 deployment, and models would likely require recalibration or retraining for new settings.

385 Second, while we implemented strict timestamp-based filtering to prevent temporal leakage, residual
 386 leakage from undocumented future events cannot be entirely ruled out. The absence of medication
 387 or diagnosis records was treated as true zero, which may introduce bias if missing records reflect
 388 documentation gaps rather than true absence.

389 Third, important operational predictors of ED-LOS (boarding time, bed availability, staffing, laboratory
 390 turnaround time, imaging delays) are unavailable in our dataset. These factors significantly influence
 391 actual ED-LOS and represent an important direction for future work.

392 Fourth, outcome definitions (e.g., admission including observation vs. inpatient only) may vary across
 393 settings. ED-LOS and hospital LOS calculations depend on accurate timestamp recording, and our data
 394 may have recording inconsistencies.

395 Fifth, class imbalance required task-specific weighting, and the modest discrimination of the ED-LOS
 396 models (AUC 0.659) and hospital LOS models (AUC 0.670) limits standalone clinical utility. These
 397 models are intended to augment, not replace, clinical judgment.

398 Sixth, calibration analysis revealed slopes deviating from 1.0, indicating that raw probabilities
 399 require recalibration before clinical deployment. The retrospective design also cannot capture clinician
 400 acceptance, workflow integration challenges, or alert fatigue [64, 65].

401 Seventh, while our fairness analysis showed relatively equitable performance across demographic
 402 subgroups, the sample sizes for some race groups (e.g., Black, $n=591$) limit the precision of these
 403 estimates. Future prospective validation should include granular calibration analysis by insurance type
 404 to fully rule out socioeconomic bias.

405 Finally, this study represents a hypothesis-generating analysis. Prospective implementation studies are
 406 essential to confirm real-world performance, evaluate clinician acceptance, workflow integration, and alert
 407 fatigue.

5 CONCLUSION

408 This study demonstrates that machine learning using data available at triage can effectively predict hospital
 409 admission (AUC 0.816, strong discrimination) and modestly predict prolonged ED length of stay (AUC
 410 0.659) and prolonged hospitalization (AUC 0.670) among admitted patients. Using the MC-MED dataset
 411 of 118,385 adult ED visits, XGBoost consistently performed best for ED-LOS and admission prediction,
 412 while Random Forest achieved the best performance for hospital LOS prediction.

413 SHAP analysis revealed that triage acuity level, age, and comorbidity burden were consistently
 414 important across all three tasks. Beyond these common predictors, EMS arrival was distinctively
 415 associated with prolonged ED length of stay, triage heart rate was strongly associated with admission
 416 decisions, and vital sign abnormalities were important predictors of prolonged hospitalization. These
 417 findings highlight how different clinical factors become relevant at different stages of patient care.

418 The admission model shows particular promise for early disposition support, while the ED-LOS and
 419 hospital LOS models provide complementary operational intelligence. However, all models require
 420 prospective validation, probability recalibration, and rigorous evaluation of clinical workflow integration

before deployment. These findings should be interpreted as hypothesis-generating for future prospective studies rather than establishing established clinical benefits. External validation across diverse healthcare settings is necessary, and models would require recalibration for new populations. This study provides an evidence base for future prospective evaluations of machine learning-based clinical decision support in emergency departments to improve patient flow and optimize resource allocation.

6 CONFLICT OF INTEREST

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

7 AUTHOR CONTRIBUTIONS

Onyedikachi I. Onwurah designed the study, implemented all machine learning models, performed the data analysis, and drafted the manuscript. Sol Richardson contributed to study design, data interpretation, and critical revision. Wu Sheng served as clinical reviewer. Xia Tong supervised the research, guided the methodological approach, reviewed all versions of the manuscript, and approved the final submission.

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9 ETHICS STATEMENT

The Stanford University Institutional Review Board approved the original data collection for the MC-MED dataset with a waiver of informed consent [57]. This study involves secondary analysis of the publicly available, de-identified MC-MED dataset, which is exempt from additional IRB review under 45 CFR 46.104(d)(4) as research involving the secondary use of existing, de-identified data.

10 DATA AVAILABILITY STATEMENT

The MC-MED dataset is publicly available on PhysioNet at <https://physionet.org/content/mc-med/1.0.1/> [57]. Analysis code is available at <https://github.com/Drglazizzo/ED-visit-12ML-prediction>.

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12 PUBLISHER'S NOTE

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Table 2. Variables available at or before triage with derivation methods, N=29

Feature Name	Data Nature or Derivation
Demographics (7)	
Age	Direct from EHR, range 18–90 years (ages >90 capped at 90 per de-identification)
Sex (Male)	Direct from EHR, binary
Race (White)	Direct from EHR, binary
Race (Black)	Direct from EHR, binary
Hispanic Ethnicity	Direct from EHR, binary
Medicare Insurance	Direct from EHR, binary
Medicaid Insurance	Direct from EHR, binary
Triage Vitals and Acuity (7)	
Triage Temperature	Direct from EHR, degrees Celsius
Triage Heart Rate	Direct from EHR, beats per minute
Triage Respiratory Rate	Direct from EHR, breaths per minute
Triage SpO2	Direct from EHR, percentage
Triage Systolic Blood Pressure	Direct from EHR, mmHg
Triage Diastolic Blood Pressure	Direct from EHR, mmHg
Triage Acuity Level	Derived by extracting numeric prefix from strings “1-Resuscitation” through “5-Non-Urgent”, range 1 to 5
Arrival Context (3)	
Arrival Hour	Derived from Arrival_time timestamp, range 0 to 23
Weekend Arrival	Derived from Arrival_time, binary indicator for Saturday or Sunday
EMS Arrival	Direct from EHR, binary indicator for Means_of_arrival equal to “EMS”
Chief Complaint (7)	
Complaint Count	Derived as number of presenting complaints from the chief complaint string
CC: Abdominal Pain	Derived by keyword matching for “ABDOMINAL PAIN”
CC: Chest Pain	Derived by keyword matching for “CHEST PAIN”
CC: Shortness of Breath	Derived by keyword matching for “SHORTNESS OF BREATH”, “DYSPNEA”, “SOB”
CC: Fall	Derived by keyword matching for “FALL”, “FELL”, “FALLS”
CC: Fever	Derived by keyword matching for “FEVER”, “TEMP”, “HYPERTHERMIA”
CC: Critical	Derived by keyword matching for suicide, overdose, trauma, bleeding, seizure, stroke, cardiac arrest, respiratory failure
Comorbidity Burden (2)	
Charlson Comorbidity Index	Derived using the Quan et al. (2005) ICD-10 mapping algorithm from pre-arrival diagnoses, range 0 to 37
Disease Condition Count	Derived as unique Clinical Classifications Software categories per patient from pre-arrival ICD-10 codes
Home Medications (3)	
Home Medication Count	Derived as count of unique medication entries per MRN from meds.csv with Entry_date ≤ index Arrival_time
Unique Medication Classes Count	Derived as distinct Med_class categories per MRN from pre-arrival medications
Polypharmacy Flag	Derived as binary indicator for Home Medication Count ≥ 5

Table 3. Hyperparameter search spaces and optimal values for Bayesian optimization (task-specific shown)

Model	Hyperparameter	Search Range	Optimal Value	Task
XGBoost	max depth	3 to 10	6	All
	learning rate	0.01 to 0.3	0.05	All
	n estimators	100 to 500	500	All
	subsample	0.6 to 1.0	0.8	All
	colsample bytree	—	0.8	All
	scale pos weight	—	4.26	ED-LOS $\geq 8h$
	scale pos weight	—	1.49	Hospital Admission
	scale pos weight	—	5.04	Hosp-LOS $> 7d$
LightGBM	num leaves	10 to 100	31	All
	learning rate	0.01 to 0.3	0.05	All
	n estimators	100 to 500	500	All
	subsample	0.6 to 1.0	0.8	All
	colsample bytree	0.6 to 1.0	0.8	All
	reg alpha	—	0.01	All
	reg lambda	—	0.01	All
Random Forest	n estimators	100 to 500	500	All
	max depth	5 to 20	15	All
	min samples split	2 to 20	20	All
	min samples leaf	—	10	All
	max features	sqrt	sqrt	All
TabNet	n steps	1 to 5	5	All
	n d	8 to 64	64	All
	n a	8 to 64	64	All
	gamma	1.0 to 2.0	1.5	All
	lambda sparse	—	1e-4	All

Table 4. Evaluation metrics for the four ML algorithms on the test set with 95% confidence intervals

Task	Algorithm	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F1-score	AUC-ROC
ED-LOS ≥ 8 h	XGBoost	56.2	29.9	71.6	51.8	0.422	0.659 [0.648–0.670]
	LightGBM	54.1	28.5	75.4	49.8	0.416	0.648 [0.637–0.659]
	RF	56.0	29.8	65.1	55.3	0.422	0.660 [0.649–0.671]
	TabNet	55.3	29.2	60.2	56.4	0.411	0.646 [0.635–0.657]
Hospital Admission	XGBoost	67.0	63.9	62.6	70.6	0.632	0.816 [0.809–0.824]
	LightGBM	66.9	63.8	62.6	70.5	0.632	0.817 [0.810–0.825]
	RF	65.0	62.5	58.1	73.7	0.684	0.812 [0.804–0.820]
	TabNet	66.6	63.6	64.2	67.5	0.695	0.811 [0.803–0.819]
Hosp-LOS > 7 d*	XGBoost	57.6	26.0	69.8	53.3	0.387	0.655 [0.646–0.664]
	LightGBM	58.7	27.1	58.0	63.4	0.377	0.647 [0.638–0.656]
	RF	70.2	30.2	62.3	72.0	0.407	0.670 [0.662–0.680]
	TabNet	61.8	28.6	54.3	68.4	0.383	0.651 [0.642–0.660]

*Among admitted patients only (test set n=4,765). Metrics are reported at thresholds selected by maximizing Youden's J on the validation set (ED-LOS: 0.421; Admission: 0.398; Hosp-LOS: 0.412).

Table 5. Subgroup sensitivity analysis results across all three prediction tasks

Subgroup	AUC-ROC (95% CI)		
	ED-LOS ≥ 8 h	Hospital Admission	Hosp-LOS > 7 o
Chief Complaint (not mutually exclusive)*			
Abdominal Pain	0.622 [0.601–0.643] (n=1,619)	0.762 [0.741–0.783] (n=1,619)	0.682 [0.653–0.711] (n=1,619)
Chest Pain	0.679 [0.654–0.704] (n=926)	0.807 [0.786–0.828] (n=926)	0.661 [0.626–0.696] (n=926)
Critical	0.657 [0.634–0.680] (n=1,377)	0.804 [0.785–0.823] (n=1,377)	0.667 [0.638–0.696] (n=1,377)
Fall	0.606 [0.578–0.634] (n=688)	0.779 [0.754–0.804] (n=688)	0.664 [0.630–0.698] (n=688)
Fever	0.643 [0.610–0.676] (n=410)	0.862 [0.838–0.886] (n=410)	0.629 [0.578–0.680] (n=410)
Other/Unspecified	0.675 [0.666–0.684] (n=7,466)	0.821 [0.813–0.829] (n=7,466)	0.677 [0.665–0.689] (n=7,466)
Shortness of Breath	0.615 [0.591–0.639] (n=658)	0.824 [0.802–0.846] (n=658)	0.686 [0.650–0.722] (n=658)
Age Group			
<65 years	0.688 [0.677–0.699] (n=8,058)	0.805 [0.795–0.815] (n=8,058)	0.696 [0.683–0.709] (n=8,058)
65–74 years	0.613 [0.594–0.632] (n=1,739)	0.762 [0.743–0.781] (n=1,739)	0.664 [0.641–0.687] (n=1,739)
75–84 years	0.598 [0.574–0.622] (n=1,374)	0.718 [0.695–0.741] (n=1,374)	0.669 [0.645–0.693] (n=1,374)
≥ 85 years	0.582 [0.553–0.611] (n=849)	0.704 [0.677–0.731] (n=849)	0.613 [0.584–0.642] (n=849)
Triage Acuity			
Acuity 1 (Resuscitation)	0.541 [0.460–0.622] (n=95)	0.770 [0.688–0.852] (n=95)	0.693 [0.573–0.813] (n=95)
Acuity 2 (Emergent)	0.610 [0.595–0.625] (n=3,180)	0.774 [0.761–0.787] (n=3,180)	0.645 [0.626–0.664] (n=3,180)
Acuity 3 (Urgent)	0.650 [0.640–0.660] (n=7,510)	0.757 [0.747–0.767] (n=7,510)	0.684 [0.672–0.696] (n=7,510)
Acuity 4–5 (Semi/Non-urgent)	0.714 [0.691–0.737] (n=1,235)	0.729 [0.703–0.755] (n=1,235)	0.719 [0.685–0.753] (n=1,235)

*Chief complaint subgroups are not mutually exclusive; patients with multiple complaints appear in each relevant subgroup.

*Small sample size (n=33) warrants cautious interpretation.